

RISING IT – ANPR Software

On-Premises MVP Product Plan

Objective: Build a simple, affordable and easy-to-deploy ANPR solution running entirely at the customer's premises. Multi-tenant/cloud functionality will be a later phase.

1. Product Scope – Version 1

Module	V1 Requirement
Installation	On-premises server/PC inside customer LAN
Cameras	Start with 1–4 ANPR cameras; architecture should allow expansion
Recognition	Receive plate number and event data from ANPR camera
Vehicle Database	Vehicle number, owner/company, type, status, validity and remarks
Whitelist	Authorized vehicles and optional automatic gate opening
Blacklist	Unauthorized vehicles, alert and event recording
Live Events	Latest events with image, plate, camera, time and direction
Search	Plate, date/time, camera and IN/OUT filters
Reports	Daily/monthly movement reports; CSV/Excel export
Users	Admin and Operator roles
Backup	Local database and image backup/restore
Integration	Barrier/relay/access controller through API or LAN I/O

2. Recommended On-Premises Architecture

ANPR Camera(s) → LAN/PoE → Network Switch → ANPR Application Server → PostgreSQL Database + Local Image Storage → Web Dashboard

Optional: Rules Engine → Barrier/Gate Controller

3. Suggested Software Stack

Layer	Recommended Technology	Reason
Frontend	React or Vue	Responsive browser dashboard
Backend	Python + FastAPI	REST API and future AI integration
Database	PostgreSQL	Reliable production database
Image Storage	Local filesystem + database references	Keeps database manageable
API	REST + WebSocket	Integration and real-time events
Deployment	Windows or Linux	Flexible customer installation

4. Main Application Screens

Screen	Main Functions
Login	Username, password and role-based access
Dashboard	Today's entries/exits, latest events, alerts and camera status
Live Events	Real-time events with plate, image, time, camera and direction
Cameras	Add/edit camera, IP, gate/location, direction and status
Vehicles	Vehicle details, whitelist/blacklist and validity
Vehicle History	Complete movement history for selected plate
Alerts	Blacklist/unknown events and operator acknowledgement
Reports	Date-wise, vehicle-wise, camera-wise and IN/OUT reports
Users	Admin/operator management
Settings	Site, backup, storage, retention and integration settings

5. Core Event Processing

Step	Action
1	ANPR camera detects vehicle
2	Camera sends plate/event to server
3	Server validates camera and event
4	Normalize registration number
5	Save plate, timestamp, camera, direction, confidence and image
6	Check whitelist / blacklist / unknown
7	Apply gate rule
8	Update dashboard and alerts

6. Initial Database Design

Table	Important Fields
users	id, username, password_hash, role, status
cameras	id, name, ip_address, gate, direction, protocol, status
vehicles	id, plate_number, owner, vehicle_type, status, valid_from, valid_to
anpr_events	id, plate_number, camera_id, event_time, direction, confidence, image_path, status
alerts	id, event_id, alert_type, message, acknowledged_by, acknowledged_at
gate_actions	id, event_id, action, result, timestamp
settings	key, value

7. Starter Hardware

Component	Starter Recommendation
ANPR Camera	1–4 IP ANPR cameras with event/API support
Server	Intel Core i5/i7, 16 GB RAM, 500 GB–1 TB SSD
OS	Windows 11 Pro / Windows Server / Ubuntu Linux
Network	Gigabit Ethernet; PoE switch where applicable
Storage	Additional HDD/NAS for long-term image retention
UPS	Recommended for server, switch and gate controller

8. MVP Development Phases

Phase	Deliverable
1	Requirements, UI wireframes and technical architecture
2	Database, authentication and camera management
3	ANPR event receiver and event storage

Phase	Deliverable
4	Dashboard, live events and vehicle database
5	Whitelist/blacklist and alerts
6	Reports, search and CSV/Excel export
7	Barrier/gate integration
8	Testing, installer, backup/restore and documentation

9. Design Principles

- No cloud dependency in V1; the system should work when Internet is unavailable.
- Browser-based interface for authorized PCs on the customer LAN.
- Camera-independent architecture using standard APIs/protocols where possible.
- Store images outside the main database and keep only metadata/path in PostgreSQL.
- Configurable event/image retention.
- Design API and database so a future cloud/multi-tenant version can reuse the core engine.
- Use local licensing in V1; evolve to cloud licensing later.

Recommended V1 target: 1–4 ANPR cameras, one on-premises server, web dashboard, vehicle database, whitelist/blacklist, event history, reports and optional barrier integration. Focus the first release on reliability and ease of installation.